

ASSIGNMENT NO.8: Views and Data Abstraction

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 4 course records and 10 student records.

USE CollegeDB;

-- =====

-- Q1) CREATE TABLES

-- =====

```
CREATE TABLE Course_View (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50) NOT NULL,  
    Fees DECIMAL(10,2) NOT NULL  
);
```

```
CREATE TABLE Student_View (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,
```

```
CourseID INT,  
City VARCHAR(30),  
Marks INT,  
Email VARCHAR(60),  
FOREIGN KEY (CourseID) REFERENCES Course_View(CourseID)  
);
```

-- Insert 4 Course records

```
INSERT INTO Course_View  
(CourseID, CourseName, Fees)  
VALUES  
(101, 'BSc IT', 45000.00),  
(102, 'BCA', 40000.00),  
(103, 'BSc CS', 42000.00),  
(104, 'Data Science', 55000.00);
```

-- Insert 10 Student records

```
INSERT INTO Student_View  
(StudentID, StudentName, CourseID, City, Marks, Email)  
VALUES  
(1, 'Rahul Sharma', 101, 'Mumbai', 78, 'rahul@gmail.com'),  
(2, 'Priya Patil', 102, 'Pune', 85, 'priya@gmail.com'),  
(3, 'Aman Verma', 103, 'Nashik', 67, 'aman@gmail.com'),  
(4, 'Sneha Joshi', 101, 'Mumbai', 91, 'sneha@gmail.com'),  
(5, 'Karan Mehta', 102, 'Thane', 74, 'karan@gmail.com'),  
(6, 'Neha Shah', 101, 'Pune', 88, 'neha@gmail.com'),  
(7, 'Rohan Desai', 103, 'Mumbai', 69, 'rohan@gmail.com'),
```

(8, 'Meera Kulkarni', 102, 'Nashik', 82, 'meera@gmail.com'),

(9, 'Arjun Nair', 101, 'Thane', 59, 'arjun@gmail.com'),

(10, 'Divya More', 102, 'Mumbai', 93, 'divya@gmail.com');

```
SELECT * FROM Course_View;
```

```
SELECT * FROM Student_View;
```

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The 'Table: Student_View' tab is active. The SQL query box contains 'SELECT * FROM Course_View;'. The result shows 4 rows of course data.

	CourseID	CourseName	Fees
☐	101	BSc IT	45000.00
☐	102	BCA	40000.00
☐	103	BSc CS	42000.00
☐	104	Data Science	55000.00

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The 'Table: Student_View' tab is active. The SQL query box contains 'SELECT * FROM Student_View;'. The result shows 10 rows of student data.

	StudentID	StudentName	CourseID	City	Marks	Email
☐	1	Rahul Sharma	101	Mumbai	78	rahul@gmail.com
☐	2	Priya Patil	102	Pune	85	priya@gmail.com
☐	3	Aman Verma	103	Nashik	67	aman@gmail.com
☐	4	Sneha Joshi	101	Mumbai	91	sneha@gmail.com
☐	5	Karan Mehta	102	Thane	74	karan@gmail.com
☐	6	Neha Shah	101	Pune	88	neha@gmail.com
☐	7	Rohan Desai	103	Mumbai	69	rohan@gmail.com
☐	8	Meera Kulkarni	102	Nashik	82	meera@gmail.com
☐	9	Arjun Nair	101	Thane	59	arjun@gmail.com
☐	10	Divya More	102	Mumbai	93	divya@gmail.com

Q2) Create a view named StudentBasicView to display only selected student details.

The view should contain:

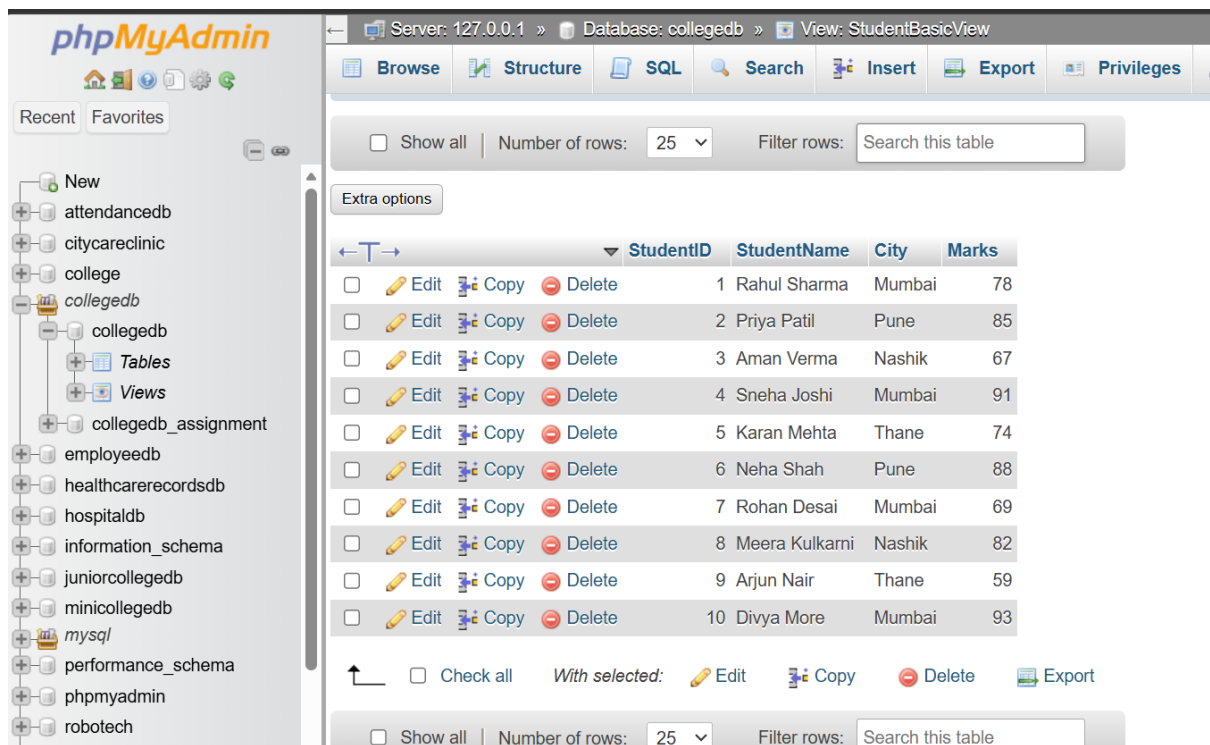
Column
StudentID
StudentName

Column
City
Marks

```
CREATE VIEW StudentBasicView AS  
SELECT StudentID, StudentName, City, Marks  
FROM Student_View;
```

-- Display the view

```
SELECT * FROM StudentBasicView;
```



Q3) Create a view named StudentCourseView using Student and Course tables.

The view should display:

Column
StudentID
StudentName
City
Marks
CourseName
Fees

CREATE VIEW StudentCourseView AS

SELECT

Student_View.StudentID,
 Student_View.StudentName,
 Student_View.City,
 Student_View.Marks,
 Course_View.CourseName,
 Course_View.Fees

```

FROM Student_View
INNER JOIN Course_View
ON Student_View.CourseID = Course_View.CourseID;

-- Display the view

SELECT * FROM StudentCourseView;

```

	StudentID	StudentName	City	Marks	CourseName	Fees
☐	1	Rahul Sharma	Mumbai	78	BSc IT	45000.00
☐	4	Sneha Joshi	Mumbai	91	BSc IT	45000.00
☐	6	Neha Shah	Pune	88	BSc IT	45000.00
☐	9	Arjun Nair	Thane	59	BSc IT	45000.00
☐	2	Priya Patil	Pune	85	BCA	40000.00
☐	5	Karan Mehta	Thane	74	BCA	40000.00
☐	8	Meera Kulkarni	Nashik	82	BCA	40000.00
☐	10	Divya More	Mumbai	93	BCA	40000.00
☐	3	Aman Verma	Nashik	67	BSc CS	42000.00
☐	7	Rohan Desai	Mumbai	69	BSc CS	42000.00

Q4) Write a MySQL query using StudentCourseView to display students from Mumbai.

The output should contain:

Column
StudentName
City
CourseName
Marks

```

SELECT StudentName, City, CourseName, Marks
FROM StudentCourseView
WHERE City = 'Mumbai';

```

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'collegedb' selected. The main panel shows a SQL query: `SELECT StudentName, City, CourseName, Marks FROM StudentCourseView WHERE City =`. Below the query, the results are displayed in a table with 4 rows. The table has columns: StudentName, City, CourseName, and Marks. The data rows are: Rahul Sharma (Mumbai, BSc IT, 78), Sneha Joshi (Mumbai, BSc IT, 91), Divya More (Mumbai, BCA, 93), and Rohan Desai (Mumbai, BSc CS, 69). The interface also shows a warning message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete'.

StudentName	City	CourseName	Marks
Rahul Sharma	Mumbai	BSc IT	78
Sneha Joshi	Mumbai	BSc IT	91
Divya More	Mumbai	BCA	93
Rohan Desai	Mumbai	BSc CS	69

Q5) Write a MySQL query using StudentCourseView to display students sorted by marks in descending order.

The output should contain:

Column
StudentName
CourseName
Marks

```
SELECT StudentName, CourseName, Marks
FROM StudentCourseView
ORDER BY Marks DESC;
```

Server: 127.0.0.1 » Database: collegedb » View: StudentCourseView

Showing rows 0 - 9 (10 total, Query took 0.0009 seconds.) [Marks: 93... - 59...]

```
SELECT StudentName, CourseName, Marks FROM StudentCourseView ORDER BY Marks DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

StudentName	CourseName	Marks
Divya More	BCA	93
Sneha Joshi	BSc IT	91
Neha Shah	BSc IT	88
Priya Patil	BCA	85
Meera Kulkarni	BCA	82
Rahul Sharma	BSc IT	78
Karan Mehta	BCA	74
Rohan Desai	BSc CS	69
Aman Verma	BSc CS	67
Arjun Nair	BSc IT	59

Q6) Create a report view named ManagementReportView.

The view should display course-wise summary:

Column
CourseName
TotalStudents
AverageMarks
LowestMarks
HighestMarks

CREATE VIEW ManagementReportView AS

SELECT

Course_View.CourseName,

COUNT(Student_View.StudentID) AS TotalStudents,

AVG(Student_View.Marks) AS AverageMarks,

MIN(Student_View.Marks) AS LowestMarks,

MAX(Student_View.Marks) AS HighestMarks

FROM Course_View

```

LEFT JOIN Student_View
ON Course_View.CourseID = Student_View.CourseID

GROUP BY Course_View.CourseName;

```

-- Display Management Report

```
SELECT * FROM ManagementReportView;
```

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'collegedb' selected. The main panel displays the 'ManagementReportView' table. A message at the top states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features'. Below this, a green status bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0025 seconds.)'. The SQL query 'SELECT * FROM ManagementReportView;' is shown in the editor. Below the query are options for 'Profiling' and 'Show all'. The table data is displayed below the query editor.

CourseName	TotalStudents	AverageMarks	LowestMarks	HighestMarks
BCA	4	83.5000	74	93
BSc CS	2	68.0000	67	69
BSc IT	4	79.0000	59	91
Data Science	0	NULL	NULL	NULL

Explanation

ManagementReportView is useful for reporting because it provides a course-wise summary of student performance in a simple and organized format.